AVIKAM JHA

Physical Design & Implementation Engineer
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SUMMARY

Computer Engineering M.S. candidate (**GPA 4.0**, Dec 2026) focused on **physical design and ASIC implementation**. Has taken two designs from RTL to GDS with Synopsys Design Compiler and Cadence Innovus, closing a **16,590-cell** block with **zero setup and hold violations across 2,196 paths** and clean DRC/LVS in Virtuoso. Also hand-lays standard cells, runs parasitic extraction, sizes transistor-level circuits for power, performance, and area, and designs the RTL that feeds the flow. Brings **2+ years as a software engineer** automating Linux infrastructure with Python and Shell.

EDUCATION

Arizona State University, Tempe, AZ | M.S., Computer Engineering | GPA 4.0/4.0 | Dec 2026
Relevant coursework: VLSI Design, Digital Systems and Circuits, SoC Design, Computer Architecture II, Digital Verification and Test

National Institute of Technology, Delhi | B.E., Electronics and Communication Engineering | July 2022

TECHNICAL SKILLS

Physical Design & Signoff Tools: Cadence Innovus, Cadence Virtuoso, Synopsys Design Compiler, Siemens Calibre (DRC, LVS, xACT 3D PEX), Synopsys IC Validator, Synopsys HSPICE, Custom WaveView

Simulation Tools: ModelSim, Xilinx Vivado (simulation)

Scripting & Languages: Tcl, Python, Cadence SKILL, Shell/Bash, Make, Verilog, SystemVerilog, C, C++

Platforms & PDKs: Linux, Git, Kubernetes; ASAP7 7nm FinFET, SAED 32nm

PROJECTS

Sparse GCN Accelerator Block, RTL to GDS | Design Compiler, Innovus, Virtuoso, ASAP7 7nm | 2025

- Implemented a **16,590-cell, 39,126-gate** block end to end: synthesis, floorplanning, power stripes, CTS, placement, routing, and post-route optimization, closing at **54.3%** placement density in **0.054 mm²** of core area.
- Closed timing with **zero setup and hold violations across 2,196 paths** (setup WNS +0.094 ns, hold WNS +0.002 ns, TNS 0) and zero max-cap, max-transition, and max-length violations after post-route optimization.
- Built a clock tree of **19 buffers and 1 inverter** over **6,294 um** of clock wirelength and reviewed the Innovus CTS report for skew and insertion delay.
- Reduced power to **5.26 mW** (1.93 mW internal, 3.33 mW switching from VCD) while meeting an **88.5 ns** latency target against a 100 ns budget, by iterating micro-architecture, zero-operand early termination, and per-block clock gating.
- Signed off **zero DRC violations**, no connectivity violations, and **LVS clean** after exporting GDS into Virtuoso.

Registered Adder, Full Flow Bring-Up | Design Compiler, Innovus, Virtuoso, ASAP7 7nm | 2025

- Ran a boundary-flopped datapath through synthesis, power stripes, CTS, APR, and GDS import into Virtuoso for DRC and LVS, with inputs on the west edge and outputs on the east edge.
- Adapted the **Tcl synthesis and APR scripts** for a multi-module hierarchy, reported core and chip area, and analyzed the top three worst-case setup and best-case hold paths from post-route timing reports.

Standard Cell Layout & Parasitic Extraction (BUFx6f) | Virtuoso, Calibre, HSPICE, ASAP7 7nm | 2025

- Laid out a 6x-drive buffer to the **7.5-track** library architecture with self-drawn M2 tracks, clean DRC and LVS, and a mirrored 3x3 array passing DRC with only the expected latch-up flag.
- Ran **Calibre xACT 3D** extraction and compared pre- and post-layout results under an FO4 load: propagation delay grew from **12.9 ps to 20.1 ps** and fall time from 6.44 ps to 9.38 ps, explained with Elmore delay analysis.

FinFET Multi-Vt Device Characterization | HSPICE, ASAP7 7nm | 2025

- Characterized **SLVT, LVT, RVT, and SRAM** devices at -40 °C, 25 °C, and 120 °C with HSPICE DC sweeps and .measure, comparing drive strength and leakage to inform multi-Vt cell selection.

Full-Custom 4-bit Adder, PPA Optimization | Virtuoso, HSPICE, IC Validator, SAED 32nm | 2024

- Designed a mirror-architecture adder and reached **240 ps**, **27.43 uW**, and **23 um²** by upsizing only the final six stacked transistors 1.5x instead of the whole circuit, earning a top score on the course power-delay-area metric.

Logic Path Sizing with Logical Effort | HSPICE, Virtuoso, SAED 32nm | 2024

- Sized a three-stage NAND/NOR path by hand (NAND 2,799 nm, NOR 2,552 nm), measured **41.4 ps / 43.4 ps** in HSPICE, and built the path from standard cells in Virtuoso with clean DRC and LVS.

5-Stage Pipelined 32-bit CPU | SystemVerilog | Aug – Dec 2025

- Built a 32-bit RISC-style core from memory, register file, ALU, and PC primitives to a full pipeline with **ID/EX, EX/MEM, and MEM/WB** registers and a hazard-detection control FSM.
- Designed a **forwarding unit** with EX/MEM-over-MEM/WB priority and three independent bypass networks (ALU operands, branch comparands, jump-register targets), resolving **five stall conditions** across load-use, jump-register, and branch hazards.
- Extended the ISA with **load-linked/store-conditional** atomics and built a **CAM** for single-cycle cache tag lookup.

RTL-to-GDS Program | In Progress, Sept – Dec 2026

- Taking parameterized IP (FIFOs, arbiter, CDC synchronizers, UART) and a RISC-V RV32I SoC through synthesis, SDC constraints, place-and-route, and DRC/LVS signoff in open-source and Cadence flows (target: Dec 2026).

EXPERIENCE

Park+ (Parviom Technologies Pvt. Ltd.), Gurugram, India | SDE 2 | Mar 2022 – Jul 2024

- Owned **Linux and Kubernetes infrastructure** for a **60+ microservice** platform across networking, security, and compliance, and built **Shell and Python automation** that cut issue-response time by **20%**.

Arizona State University | Graduate Service Assistant | Aug 2025 – May 2026

- Evaluated EEE 120 digital design labs (adders, multiplexers, decoders, ALU, counters, FSMs) and graded EEE 202 circuit analysis, checking CAD and SPICE simulations for correctness and timing.

Arizona State University | Research Aide, Introduction to Materials Science | May – Aug 2025

- Authored **40+ assignment questions** and technical documentation across 10+ semiconductor topics for a 120+ student cohort.